

Dimensional Modeling

Sample Case Study

You are required to design a Dimensional Model in the way that it fulfills the requirement for the following system.

Grocery System (POS)

The following queries shall be generated through your design:

1. Total sales of a particular product from all stores in the last quarter
2. Total sales by product by store by month
3. Yearly profit generated by stores in the north region
4. How customer deviates from store to store with particular products
5. When I promote one thing how does it affect the other
6. Check if more Products are sold on 1st 10 days and 20th to 25th date of the month than the whole month.
7. Average daily sales (in dollars) of product categories.
8. The total number of customers purchasing a particular product.
9. The total number of customers visiting a particular store in a month.
10. Count how many people buy with coupon.

Design Requirements

Here is the eight points of the complete dimensional modeling design:

1. The processes, and hence the identity of the fact tables
2. The grain of each fact table.
3. The dimensions of each fact table
4. The facts, including pre-calculated facts
5. The dimension attributes with complete descriptions and proper terminology
6. How to track slowly changing dimensions
7. The historical duration of the database
8. The urgency with which the data is extracted and loaded into the data warehouse.

Grocery System (POS):

1. The processes, and hence the identity of the fact tables

Following are the major processes in the Grocery System:
Sales, inventories, cost, revenue, buyer (Customer), etc.

Identities of Fact Tables:

Sales Base Fact Table:

- i) Time_key
- ii) Product_key
- iii) Store_key

Quarterly Aggregate Fact Table:

- i) Product_key
- ii) Quarter_key
- iii) Store_key

Monthly Aggregate Fact Table:

- i) Month_key
- ii) Product_key
- iii) Store_key

Yearly Region Wise Aggregate Fact Table:

- i) Product Key
- ii) Yearly_key
- iii) Region_key

Customer Base Fact Table:

- i) Customer key
- ii) Product key
- iii) Store key
- iv) Promo key
- v) Time key

2. The grain of each fact table:

The grain of a fact table is the least level of each dimension.

Sales Base Fact Table:

The grain level is the fact measurements by day by store and by product.

Quarterly Aggregate Fact Table:

The grain level is the sales by quarter by store and by product.

Monthly Aggregate Fact Table:

The grain of this table is total sales by month by store and by product.

Yearly Region Wise Aggregate Fact Table:

The grain level of this fact table is sales by year by region and by product.

Customer Base Fact Table:

The grain level of this fact table is by day by store by product by customer and by promotion.

3. The dimensions of each fact table:**Sales Base Fact Table:**

Time, Store, Product

Quarterly Aggregate Fact Table:

Quarter, Store, Product

Yearly Region Wise Aggregate Fact Table:

Year, Region, Product

Customer Base Fact Table:

Time, Store, Product, Customer, Promotion

4. The facts, including pre-calculated facts:

The major facts and pre-calculated facts are:

- i) Quantity_sold
- ii) Dollar_profit
- iii) Cost
- iv) Average_sales
- v) Quartely_quantiy_sold
- vi) Quartely_cost
- vii) Yearly_profit
- viii) Customer_Count
- ix) Product_Count
- x) Product_Quantity_per_Customer

5. The dimension attributes with complete descriptions and proper terminology:**i) Time Dimension**

Time_key It is the identifier of the time Dimension

Day description It keeps the day info

Day_one_to_ten It keeps dates from 1st to 10th

Day_twenty_to_twentyfive It keeps dates from 20 to 25

Week_key It is the identifier of the week level of time dimension

Week Description

Month_key It is the identifier of the month level of time dimension

Month Description

Quarter Key It is the identifier of the quarter level of time dimension

Quarter Description

Year Key It is the identifier of the year level of time dimension

Year Description

ii) Store Dimension

Store_key It is the identifier of the store dimension

Store_number It keeps the store number of the store

Store_address It keeps the address of the store

City

Country

State

Zip

Manager

Phone

Fax

Floor_plan_type It stores the current floor plan type of the store

Region_key It is the identifier of the region level of store dimension

Region Description

iii) Product Dimension

Product_key it is the identifier of the product dimension

SKU_number It is the SKU number of the production

SKU_Desc It is the Description of the SKU

Package_size It keeps the size of package of the product

Brand It stores the Brand of the product

Diet It keeps the diet info of the product

Weight_type It is the weight of the product

Cases_per_pallet It keeps cases per pallet

Subcategory It keeps the subcategory of the product

Category

Department

iv) Customer Dimension:

Customer_key It is the identifier of the customer dimension

Customer_SSN It keeps the SSN no of the customer for tracking

Customer_name It keeps the name of the customer

Customer_phone_no

Customer_region

v) Promotions Dimension

Promotion_key It is the identifier of the promotion Dimension

Promotion_name It is the name of the promotion

Price_reduction_type

Ad_type It stores the advertisement type of the promotion

Display_type It stores the display type of the promotion

Cupon_type It stores the coupon types offered in the promotion

Ad_media_name It stores the media name of the advertisement

Display_provider It stores the provider of the displayer.

Prmo_cost It keeps the cost of the promotion

Promo_begin_date It keeps the start date of promotion

Promo_end_date It keeps the end date of promotion

6. How to track slowly changing dimensions:

We will track slowly changing dimensions by Type-TWO approach.

7. The historical duration of the database:

The historical duration of database is 7 to 8 years approximately however it varies according to the type of database and under certain requirements and constraints.

8. The urgency with which the data is extracted and loaded into the data warehouse:

The urgency with which the data is extracted and loaded into the data warehouse is daily.

Dimensional data Model - Grocery System (POS)

